1) water jug problem:

given a 4 gallon jug and one 3 gallon jug, with no measuring marks obtain 2 gallons of water.

Solution steps:

* Fill the 4 gallon jug completely
* Pour water from the 4 gallon jug into the 3 gallon jug.
* Now 1 jug is left in 4 gallon jug, remains in 4 gallon jug
* Empty the 3 gallon jug.
* Transfer the remaining 1 gallon into the 3 gallon jug.
* Fill the 4 gallon jug
* Pour water into the 3 gallon jug until it is full. since it already contains 1 gallon, only 2 gallons are transferred.

Result:

Exactly 2 gallons remains in the 4-gallon jug

2) Mars rover - intelligent agent.

(i) percepts:

camera images, terrain conditions, obstacles, temperature, soil composition and environmental data

(ii) operating environment:

Martian surface, including rocks, dust, craters, slopes, and extreme temperatures.

(iii) Actions:

Move, turn, take, photographs, collect samples, analyze soil, and transmit data to earth.

(iv) Performance evaluation:

Based on scientific discoveries successful sample collection, navigation accuracy energy efficiency and mission safety

(v) suitable agent architecture:

A model based goal oriented agent is suitable because it maintains information about its surroundings and plans actions to achieve scientific objectives while avoiding obstacles

5) 8- Queen problem

Problem: place 8 queens on an 8 x 8 chessboard so that no two queens attack each other.

Problem formulation:

* Initial state: An empty chessboard

* State: position of the queens placed so far.

* Actions: place a queen in a safe square.

* Goal test: All 8 queens are placed without sharing the same row column, or diagonal

* path cost: Number of placements or search steps.

Search strategies:

Backtracking, depth-first search (DFS) and hauristic search can be used.

Conclusion:

The problem is solved when all 8 queens are placed safely on the board.

9) PEAS cab booking problem

Suitable agent : goal based agent with cost based decision-making

The agent considers the pickup location, distination, cab type, fare, availability, and estimated travel time to select a suitable cab.

Pseudo code:

```
START
Input pickup_location and destination
Display available cab type
Input preferred_cab-type
Find available cabs
calculate estimated fare and travel time
select a suitable available cab
Display cab details and estimated fare
confirm booking
IF confirmed
    Book cab
    Display driver and trip
details
ELSE
    cancel booking
END IF
STOP
```

Result: The agent helps the customer choose and book a suitable cab based on their preferences.

5) uniform cost search

Given: Find the minimum-cost path from starting node S to goal node G using the weighted graph.

The edge are:

S → A = 1

S → G = 12

A → B = 3

A → C = 1

B → B = 3

C → D = 1

C → G = 2

D → G = 3

UCS procedure:

* Start at S with cost 0
* Expand S: A(1), G(12)
* Expand A(1): B(4), C(2)
* Expand C(2): D(3), G(4)
* Expand D(3): G(6), the existing route

to G with cost 4 is chosen so that it.

Select G with the lowest cost of 4 and stop

optimal path

$$S \rightarrow A \rightarrow C \rightarrow G$$

Total cost

$$1+1+2 = \boxed{4}$$

conclusion: uniform cost search find the least cost path by expanding the node with the lowest cumlative path cost first